Dafa Dega Wijaya

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Profile

Informatics Engineering undergraduate at Institut Teknologi Sepuluh Nopember (ITS) with hands-on experience in full-stack web development, backend engineering, and AI-powered applications. Experienced in developing web-based management systems, designing database architectures, building RESTful APIs, and integrating AI features into software applications. Also actively involved in technical organizations, collaborative projects, and event management. Interested in building scalable, practical, and data-driven technology solutions.

Experience

Schematics 2026 – Now

- Divison External relation
- Sub-Division Publication

School Ambasador 2021 – 2023

- MC School anniversary celebration
- Mentoring and socialization
- Welcoming external guest

FORDA(Imajaya ITS) 2025 – 2026

- Division Event
- Departemen Internal

INI LHO ITS 2026 2025 – 2026

- Sub-Division External
- Division Public Relation
- PIC & Stakeholder contact person
- Mentor & PIC Visit (Worl wide) Johor Bahru

Informatics Engineering Graduation Committee 2026

- Sub-Division External
- Divisi Liaison Officer

Basketball 2020 – 2023

- Liga PERBASI
- DBL
- SMA Awards

Organization 2019 – 2024

- OSIS
- Pramuka
- Classmeet leader

DATATHON RISTEK FASILKOM UI 2026

AVALON AI Group - (PT.Automasi Visi Analitika Optima) 2025 – Now

- Member
- Software Engineering

Bhakti Electics 2026 – Now

- house of service
- making solar panels and a community website

INI LHO ITS 2027 2026

- Sub-Division External
- Division Public Relation
- Expert Staff

DevSprints Accelerator 2026

- AI Feature Developer

WEB Development UCS 2026
(Ukm Cyber Security ITS)

- Backend Development

SKILLS

- CORE COMPETENCIES
- Leadership & Team Coordination
- Stakeholder Communication
- Public Speaking & Presentation
- Cross-functional Collaboration
- Event & Project Coordination
- Problem Solving & Adaptability

TECHNICAL SKILLS

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- Programming Languages: Python, JavaScript, TypeScript, Dart, C, C++, C#
- Frontend Development: React.js, HTML5, CSS3, Vite, Tailwind CSS
- Backend Development: Node.js, Express.js, RESTful APIs
- Database & ORM: PostgreSQL, MySQL, Supabase, Prisma ORM, IndexedDB
- AI & Machine Learning: PyTorch, OpenCV, U-Net, Semantic Segmentation
- Systems & Networking: Linux, Docker, Wireshark, Nginx, GNS3, Bash
- Software Engineering: OOP, Unit Testing, Data Modeling, Algorithm Design
- Tools: Git, GitHub, Vercel, Visual Studio Code, VMware

TRAINING

- **Boothcamp Advanced**

  Organized by schematics about the basics of programing

- **SynchroMove**

  Organized by schematics about web

- **Boothcamp Basic**

  Unvelling servo dynemics with ICHIRO

- **HACKTIVE8**

  AI for work career readiness with Google AI Product

- **NALARA Academy National AI & Deep Learning Acceleration Bootcamp(Universitas Brawijaya)**

  Level Intermediate

- **MAGE 12 ITS(**HIMATEKKOM**)**

  Workshop Multimedia With project make a AI-based website

- **DevSprint Accelerator**

  A 6-week accelerator program specifically designed for tech students. 'Code Less. Architect More. Ship in 6 Weeks'

Project

Todo App Progressive Web Application (PWA)

- Developed using JavaScript, IndexedDB, and Service Workers.
- Implemented Personal and Shared Workspaces.
- Offline functionality with local data storage.
- Outbox mechanism and automatic data synchronization when connectivity is restored.
- PWA manifest configuration and application installation support.

E-TICKET PARKING

- Designed ERD Dan SQL Queries
- Focused on transaction history and reporting

Laundry Application

Laundry Management Website
• Designed and developed a web-based laundry management system
• Implemented user authentication (admin & cashier roles)
• Features include order management, customer records, and laundry status updates
• Built using PHP, MySQL, HTML, CSS, and JavaScript
• Applied CRUD operations and basic database normalization

Warung Madura Stock Management System
Full-Stack Developer | React, TypeScript, Express.js, Prisma, PostgreSQL

- Developed a full-stack inventory and sales management application for retail operations.
- Designed and implemented RESTful APIs using Express.js and Prisma ORM.
- Built inventory management, sales transaction, and stock movement tracking features.
- Implemented data analysis capabilities to support inventory monitoring and business decision-making.

SILAJU

- An AI-powered pothole detection system that smartly prioritizes road repair spots based on the traffic density around them.

Satellite Image Segmentation (Nalara AI)Machine Learning Engineer

- Trained and evaluated a U-Net deep learning architecture to perform precise semantic segmentation on satellite imagery of forest areas.
- Optimized model parameters in Google Colab to improve boundary detection and segmentation accuracy.

RICE Milling Factory Management System

A management system that supports rice-mill operations, from purchasing paddy rice through processing, inventory management, sales, and business data analysis.

Frontend: React, TypeScript, Vite, and Tailwind CSS.

- Provides dashboards and forms for purchasing, drying, milling, inventory, and sales workflows.
- Helps users monitor operational and financial activity.

Backend: Node.js, Express, and PostgreSQL.

- Provides APIs and manages transaction data.
- Supports authentication, input validation, database migrations, and tenant-scoped data access.
- Manages purchasing, production, inventory, sales, and financial records.

AI Feature Root-Cause Analysis: React, Vite, Express, and OpenRouter.

- Defines a context contract and JSON Schema for analysis periods, metrics, symptoms, and potential drivers.
- Applies an allowlist-based privacy filter so fields outside the approved contract are excluded from AI requests.
- Includes draft SQL queries with tenant scoping through created_by.
- Includes a canary test that checks whether fake national ID numbers, phone numbers, and email addresses are removed from the payload.
- Provides a sample payload linking three days of milling-machine downtime to lower production output and revenue.
- Project focus: Build an integrated operations system and establish a secure foundation for data-driven root-cause analysis.

Web Development (Kawan-Inap & RedSeller)Frontend Developer | React.js, HTML/CSS, Vercel

- Developed interactive, client-side web applications tailored for hotel sales and commercial landing pages.
- Configured CI/CD pipelines to deploy static web applications seamlessly via Vercel and GitHub Pages.

Pokédex Application

- An app to display Pokémon information.
- Experience building a data-driven app with a user interface.

Bioskop Mini – Object-Oriented Programming

- C# dan .NET.
- Implementasi OOP.
- Polymorphism, studio, kursi, dan jadwal tayang.
- Sistem pemesanan serta pencetakan tiket.

Shadow Grid – A Stealth Game*

- Game stealth berbasis grid.
- Algoritma A* untuk NPC pathfinding.
- Sistem deteksi pemain oleh musuh.
- Line of Sight menggunakan Bresenham.
- Implementasi logika dinding, lantai, dan jarak pandang.

EDUCATION

- **SMAN 2 Lumajang** 2021 – 2024
  Math and Science
- **Institut Tekologi Sepuluh Nopemer** 2025 – 2029
  Informatics Engineering